LIGHTING DESIGN STUDY · DEMONSTRATION PROJECT

Open-plan living / kitchen



A 36 m² open-plan room, lit by 13 luminaires on five circuits. It is designed, calculated, checked against Radiance and rendered from **one set of photometric files**, so the pictures and the numbers describe the same light.

566 lx
COUNTER, FULL SCENE

66 lx
LIVING, CALM SCENE

4.3 W/m²
ALL ON

+1.4 %
BIAS VS RADIANCE

Scenes: Cook & Gather (everything on), Dinner and Sensory Calm. The last two have their dimming levels solved for zone targets. Daylight factor under a CIE overcast sky.

SCENES

Three scenes, same room, same camera



Cook & Gather — 156 W



Dinner — 32 W, dimming solved for 175 lx on the table



Sensory Calm — 31 W, no downlights, living area held in the 40–80 lx band

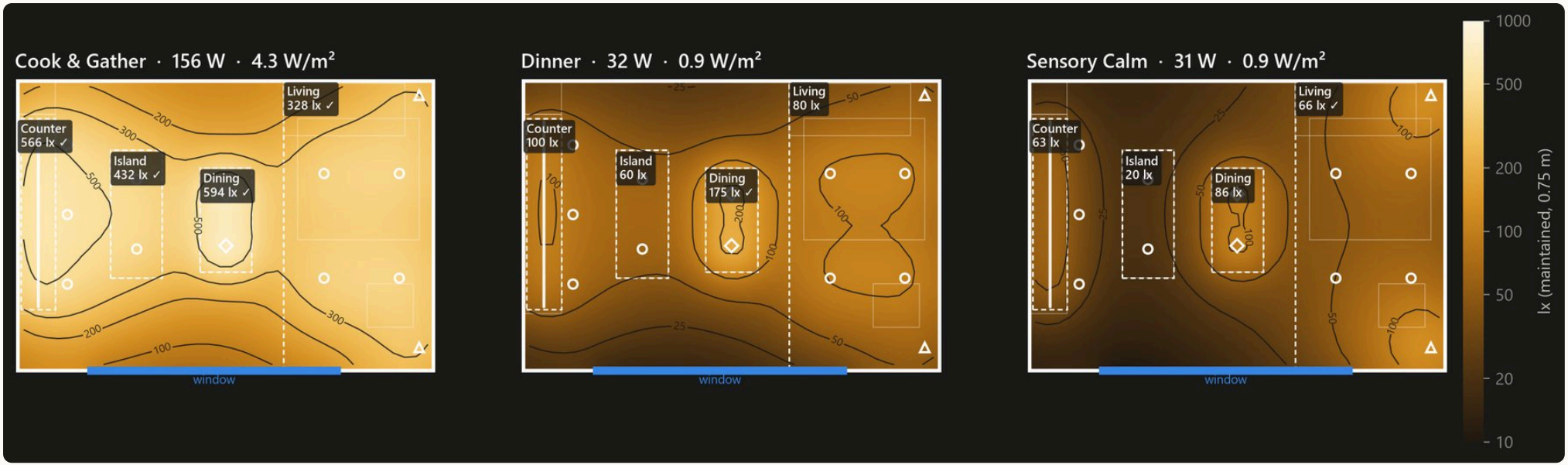


Daytime (illustrative render; daylight numbers on page 5)

Renders: Blender 5.2 Cycles. Light power comes from the IES files through a measured calibration, and exposure is noted per render.

CALCULATION

Illuminance on the 0.75 m workplane



Scene	kitchen downlights	living downlights	pendants	floor lamp	under cabinet	Power	W/m²
Cook & Gather	100 %	100 %	100 %	100 %	100 %	156 W	4.3
Dinner	11 %	30 %	34 %	13 %	24 %	32 W	0.9
Sensory Calm	0 %	0 %	15 %	63 %	21 %	31 W	0.9

Maintained values (MF 0.8). Direct light plus radiosity interreflection (2238 patches). Empty-room model.

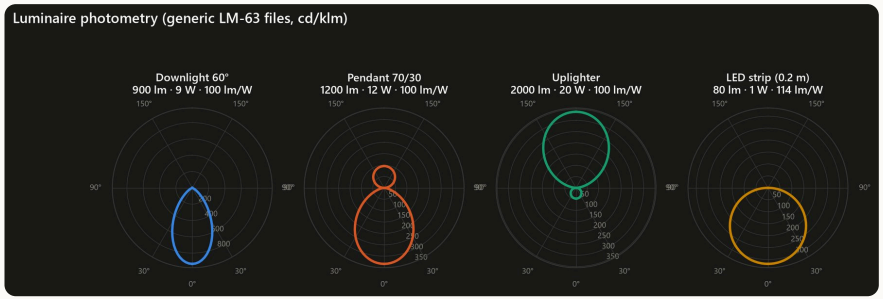
RESULTS BY ZONE

Targets, uniformity and power

Scene	Zone	Ē lx	Emin	Emax	U_0	Checks
Cook & Gather	Kitchen counter	566	331	675	0.58	✓ E_avg >= 300 lx · ✓ U0 >= 0.40
Cook & Gather	Island	432	302	508	0.70	✓ E_avg >= 300 lx · ✓ U0 >= 0.40
Cook & Gather	Dining table	594	380	759	0.64	✓ E_avg >= 200 lx · ✓ U0 >= 0.40
Cook & Gather	Living area	328	105	428	0.32	✓ E_avg >= 100 lx · ✓ U0 >= 0.30
Dinner	Kitchen counter	100	58	120	0.59	—
Dinner	Island	60	47	67	0.78	—
Dinner	Dining table	175	101	231	0.57	✓ E_avg >= 150 lx
Dinner	Living area	80	23	110	0.29	—
Sensory Calm	Kitchen counter	63	36	76	0.57	—
Sensory Calm	Island	20	17	24	0.87	—
Sensory Calm	Dining table	86	51	110	0.60	—
Sensory Calm	Living area	66	27	135	0.41	✓ 40-80 lx band (avg)

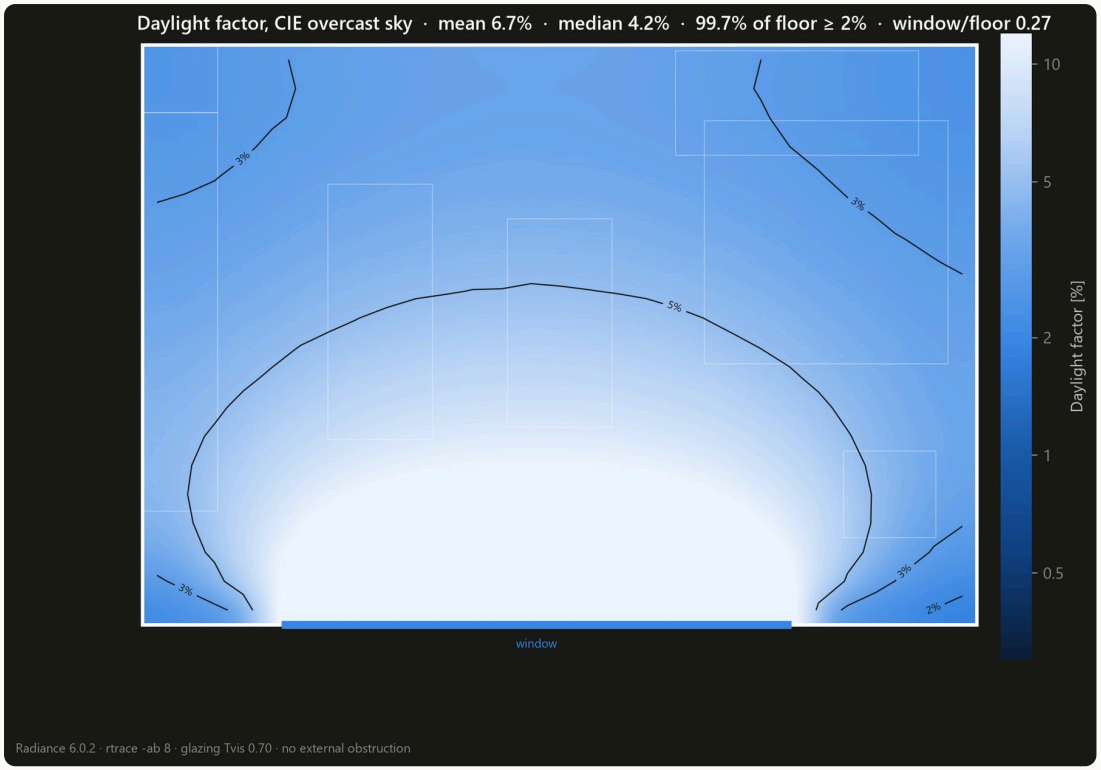
Type	Description	Qty	lm	W	lm/W	CCT
DL	Recessed downlight 60 deg, 900 lm	9	900	9.0	100	3000 K
PD	Pendant direct/indirect 70/30, 1200 lm	2	1200	12.0	100	2700 K
FL	Floor uplighter 85% up, 2000 lm	2	2000	20.0	100	2700 K
UC	LED strip segment 0.2 m, 80 lm	1	80	0.7	114	3000 K

UC is a 3.2 m LED strip modelled as 16 segments of 0.2 m.



DAYLIGHT

Daylight factor, CIE overcast sky



6.7 %

MEAN DAYLIGHT FACTOR

99.7 %

OF THE FLOOR AT DF ≥ 2 %

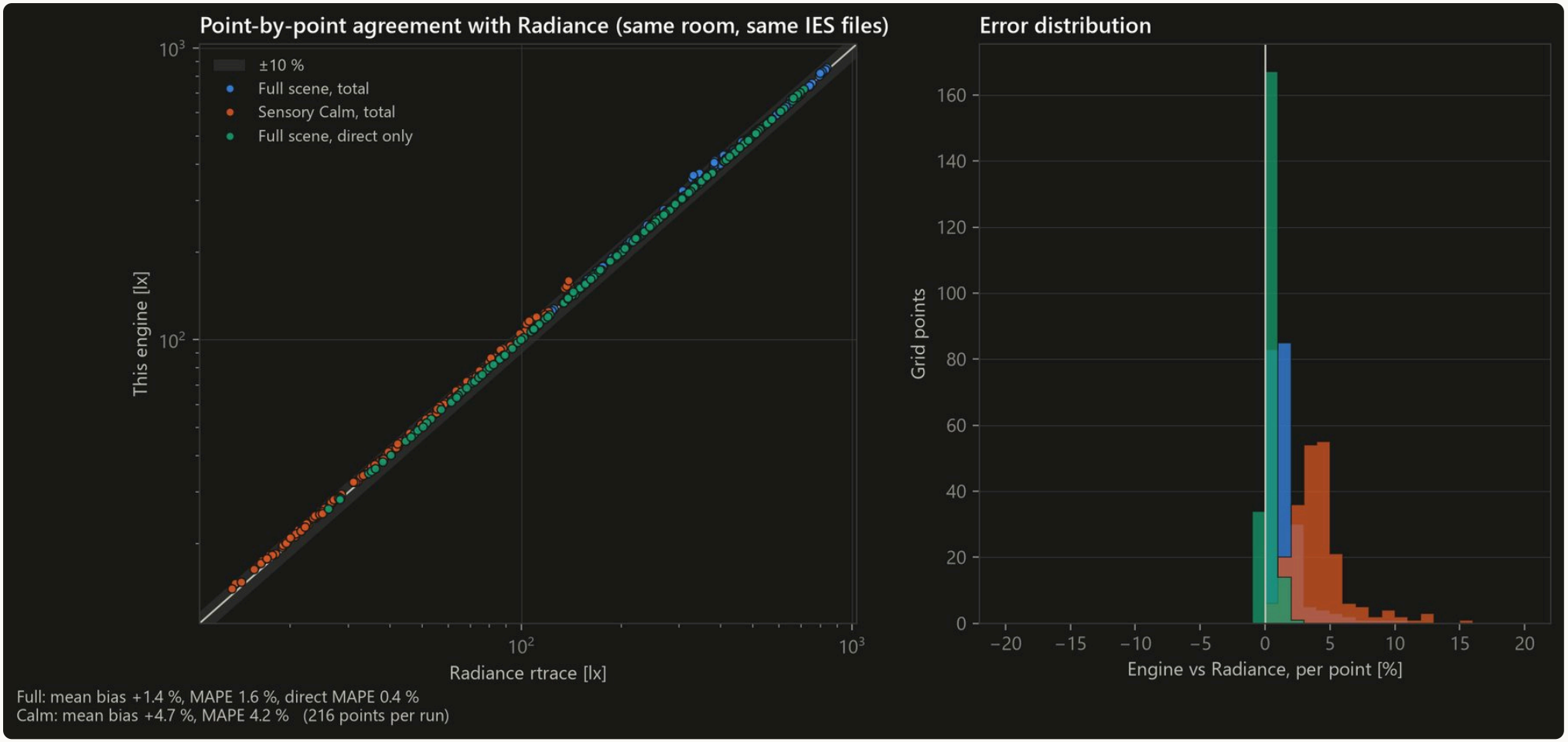
0.27

WINDOW-TO-FLOOR RATIO

The window-to-floor ratio of 0.27 falls in the 0.20–0.30 range that the author's MSc field study associated with occupant light satisfaction in Danish dwellings. The study had 13 dwellings and 242 merged observations, and it is research evidence, not a design rule.

VERIFICATION

Checked against Radiance, and the renders checked against the engine



Full scene: engine 411 lx vs Radiance 405 lx (mean bias +1.4 %, MAPE 1.6 %, direct-only MAPE 0.4 %).
Sensory Calm: mean bias +4.7 %, MAPE 4.2 %. 216 grid points per run.

Table-top illuminance read from the calibrated render passes (with furniture), compared with the empty-room engine. The render reads higher because it also contains furniture reflections and some residual blue-hour sky light through the window, which the night-time calculation leaves out:

- B full: render 639 lx vs engine 594 lx (+8 %)
- B calm: render 95 lx vs engine 86 lx (+11 %)
- B dinner: render 200 lx vs engine 175 lx (+14 %)

Automated tests: 14 closed-form and conservation checks (inverse-square/cosine law, form factors against catalogued values, enclosure and reciprocity, integrating-sphere solution, flux balance, LM-63 round trip, solver bounds). Run: `python -m pytest tests`.

Same room and same IES files in both tools. Radiance treats luminaires as finite disks, the engine as point sources.

PHOTOMETRIC RENDERS

Luminance and surface illuminance from the render passes



Luminance, cd/m² (Cook & Gather)



Illuminance on every visible surface, lx (Sensory Calm)

Limits: furniture is not in the calculation. No glare (UGR/DGP), vertical illuminance or melanopic metrics yet. The photometry is generic, so a real project needs the manufacturer's IES files. This is not a compliance certificate. Have a qualified lighting professional review any real project.